FIT5230 Malicious AI

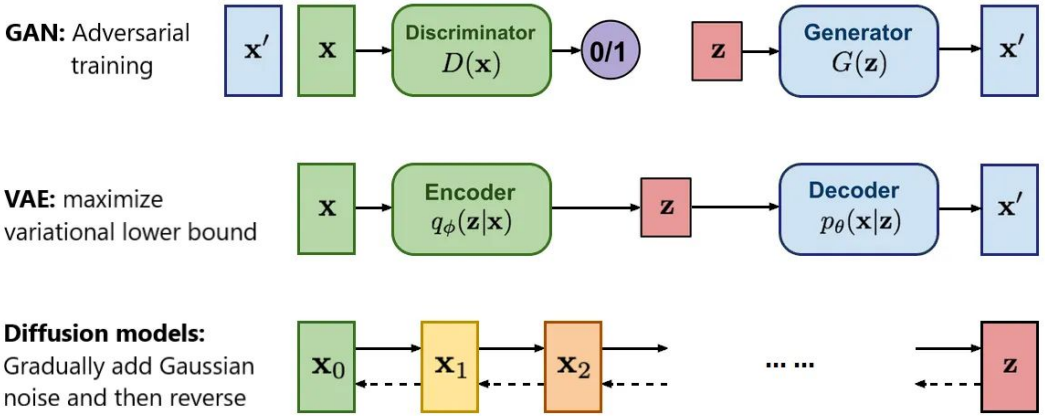
Defense vs Generative AI Fakes

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Generative AI Models
Quick Recap

Generative AI

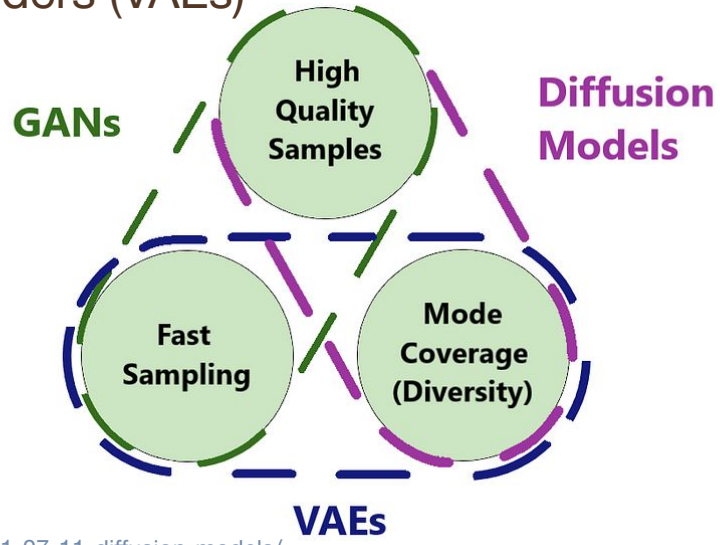
- Generative Adversarial Networks (GANs)
- Diffusion Models
- Variational Autoencoders (VAEs)



Reference: https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

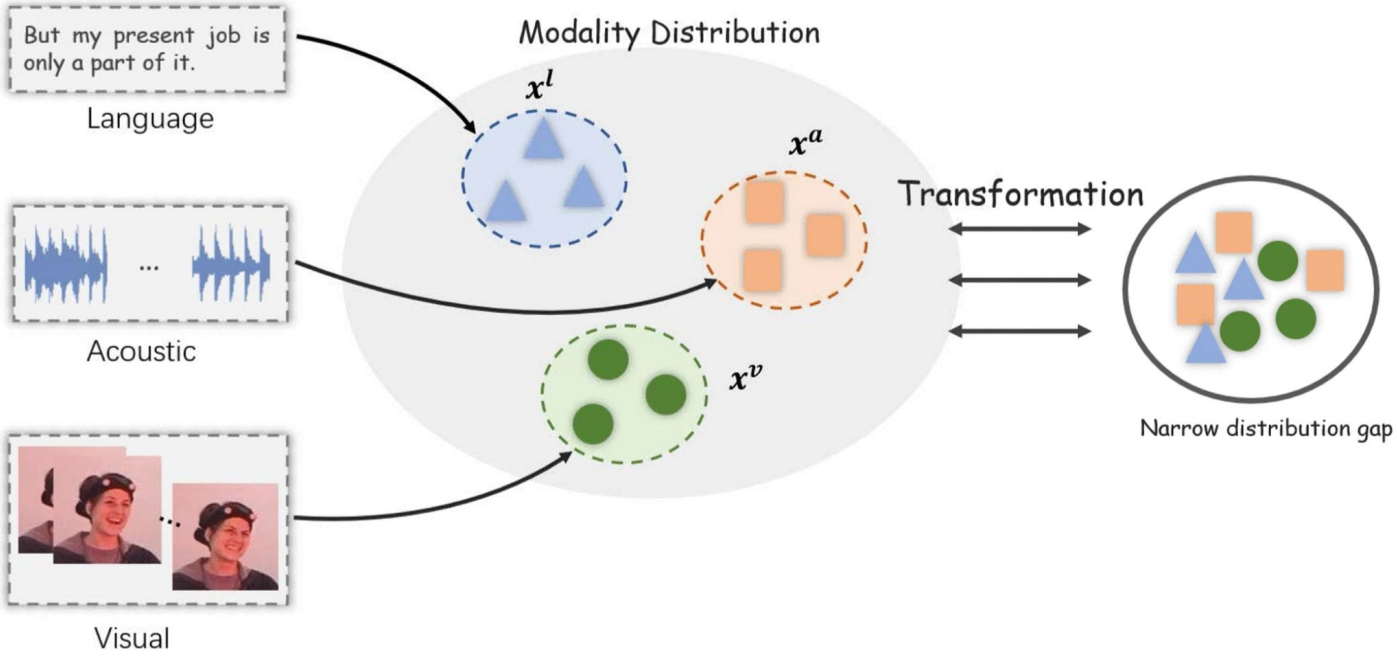
Generative AI

- Generative Adversarial Networks (GANs)
- Diffusion Models
- Variational Autoencoders (VAEs)



Reference: https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Unimodal vs Multimodal AI



Reference: Learning to Learn Better Unimodal Representations via Adaptive Multimodal Meta-Learning, IEEE Transactions on Affective Computing 2023

Unimodal vs Multimodal AI

Key Differences

Criteria	Unimodal AI	Multimodal AI
Data Input	Use single data input	Use multiple data input (e.g. image, audio or text)
Resources	Need less resources due to simple processing	Require more computational resources
Information Processing	Process a single modality, limiting the depth and accuracy of understanding	Process multiple modalities simultaneously for richer understanding
AI Models	GPT-3, BERT, ResNet	DALL-E, CLIP, SwinBERT
Application	Best suited for single type task, such as speech recognition	Works for multiple type task such as image and video captioning

Reference: https://www.index.dev/blog/comparing-unimodal-vs-multimodal-models

Unimodal vs Multimodal AI

Factors to Consider When Choosing a Modality

- Data Availability
- Task Relevance
- Encoding Complexity
- Fusion Complexity
- Performance: What performance potential could that modality provide?

A balance between these factors will help you determine the most effective modality for your specific application.

Reference: https://www.index.dev/blog/comparing-unimodal-vs-multimodal-models

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AI Security
Defense & Attack

Risk of Data Leakage

- Generative models that are trained on sensitive data might inadvertently reveal this sensitive information.
- personally identifiable information, confidential business information, or access tokens.

- E.g., inputting a dataset containing sensitive information into the prompt
- Existing LLMs may occasionally reveal private information
- Malicious users could try to extract the training data
- Disastrous for domain such as healthcare and finance

AI Phishing - Automated Phishing

- How it happens?
  - Creating hyper-personalised phishing emails
  - Tailored to each victim's likes and interests

- How does generative AI make phishing attack more dangerous?
  - More realistic → lesser grammatical mistakes, professional writing style
  - Large language models (LLMs) absorb real-time information from news outlets, corporate websites and other sources.
  - Incorporating of-the-moment details
  - AI chatbots can create and spread business email compromise and other phishing campaigns at a much faster rate than humans

AI Phishing - Spear phishing

- Has specific target individuals or organizations to steal sensitive data or deploy malware.
- Highly customized and involves extensive research on the intended victim

- How it happens?
  - Social engineering from social media sites, data breaches
  - Craft convincing messages that appear to come from a trusted source, like a colleague or friend.
  - These messages might contain malicious links, attachments to steal information

AI Phishing - Vishing

- Voice phishing – uses phone calls, voice messages and voicemails
- Trick people into sharing sensitive information

- How traditional vishing scam happens?
- Collects information on a target
- Makes a call or leaves a message pretending to be a trusted contact.
- E.g., a massive ransomware attack on MGM Resorts

AI Phishing - Vishing

How does generative AI make Vishing attack more dangerous?

- An LLM i.e., GPT-3, collect information from across the web, nearly instantly.
- Attackers can also use generative AI to clone the voice of a trusted contact and create deepfake audio.
- E.g., an employee receives a voice message from someone who sounds exactly like the CFO, requesting an urgent bank transfer.
- Read more about it here.

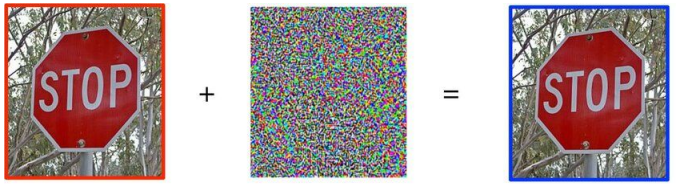
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Adversarial Attack on AI System

AI Attack

Adversarial Attacks on AI Systems

- Adversarial attacks on Generative AI
  - Create data that looks real to humans but is misleading for machine learning → adding perturbation
  - It can cause machine learning models to make ‘incorrect’ decisions.
  - Without altering the original character of the input.



Q: If hackers can deliberately inject perturbations to decision made by the machine learning model ‘incorrect’, can we still trust the decision made by the AI systems?

Reference: It Is All About Data: A Survey on the Effects of Data on Adversarial Robustness, March 2023

White-Box Attack: FGSM

White-box attacks: attacker has complete knowledge of the AI model

- Exploit the internal structure of a neural network
- Find input perturbations that lead to incorrect predictions.
- Allows attackers to use gradient-based methods
- E.g., calculate how much each pixel in an image influences the model's decision.

1. The Fast Gradient Sign Method (FGSM) involves:

- Calculating the gradient of the model's loss function with respect to the input image
- Adding a small amount of this gradient to the input to maximize the loss → effectively tricking the model

Application of FGSM

Aim of Attack: Turning a Refrigerator into a Wardrobe

- Take an image of a refrigerator and manipulate it in such a way that it is misclassified as a wardrobe.
- FGSM uses the gradient of the model's loss function to adjust the input image.

FGSM attack:

- Compute the difference between the predictions and the target label → loss value of the model
- Loss value are used to compute the gradients
- Perturbation is added to the input image by taking the sign of the gradient

Application of FGSM

Adversarial prediction

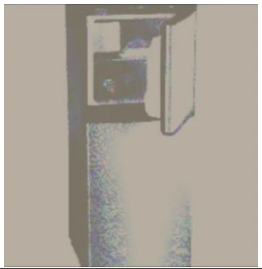
- A forward pass is performed on the adversarial image, and the model's prediction is obtained.
- The resulting label should be different from the original image's class → changed due to the perturbation

Generated output:

- The adversarial image should look like a refrigerator to human eye but should ideally fool the model into predicting the "wardrobe" class.

Original Image





Adversarial Image

Black-Box Attack

Black-Box Attacks: have limited information about the AI model.

- Might not know the model's architecture or parameters but can still craft adversarial inputs to fool it.

Black box attack with GANs:

- **Reversal** Attack: Synthesizing the original image from the hash
- **Poisoning** Attack: synthesizing hash collisions for arbitrary natural image distributions.

Black-box Attack: Idea of Hashing

- Perceptual hash is a short hexadecimal string based on an image's appearance.
  - E.g., 00081c3c3c181818
- Perceptual image hashes are not cryptographically secure hashes.
  - Aim to be smoothly invariant to small changes in the image, i.e., rotation, crop, gamma correction, noise addition, adding a border
- This is in contrast to cryptographic hash functions that are designed for non-smoothness and to change entirely if any single bit changes.

Black-box Attack: Idea of Hashing

- The perceptual hashes of the below images are only slightly changed by the text modification, but their md5 hashes are completely different.

Original



- a_hash(original) = 3c3e0e1a3a1e1e1e
- md5(original) = 8d4e3391a3bca7...

Modified



- a_hash(modified) = 3c3e0e3e3e1e1e1e
- md5(modified) = c27baa59355d10...

Reference: https://towardsdatascience.com/black-box-attacks-on-perceptual-image-hashes-with-gans-cc1be11f277

Black-box Attack: Hash Reversal

Black box attack with hash reversal attack:

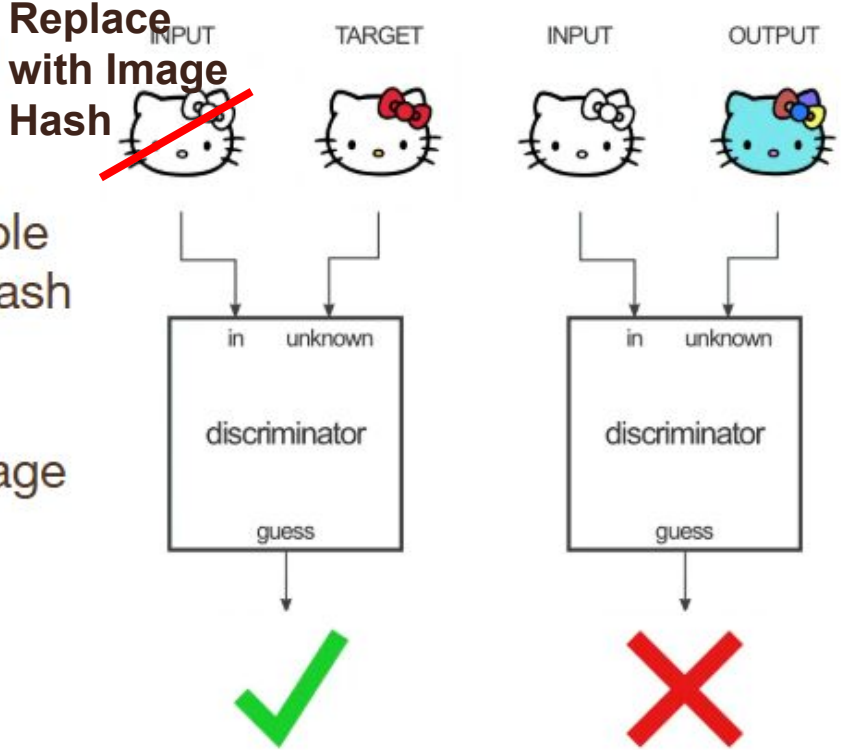
- Due to smoothness property of perceptual image hashes that relates inputs to outputs:
- this process and its inverse can be modeled with a neural network.
- There are many potential images from a single image hash
- Learn the image manifold with GAN and enable the model to explore these differing, but valid image distribution outputs.

Black-box Attack: Hash Reversal

The Pix2Pix network can be used to convert perceptual images hashes.

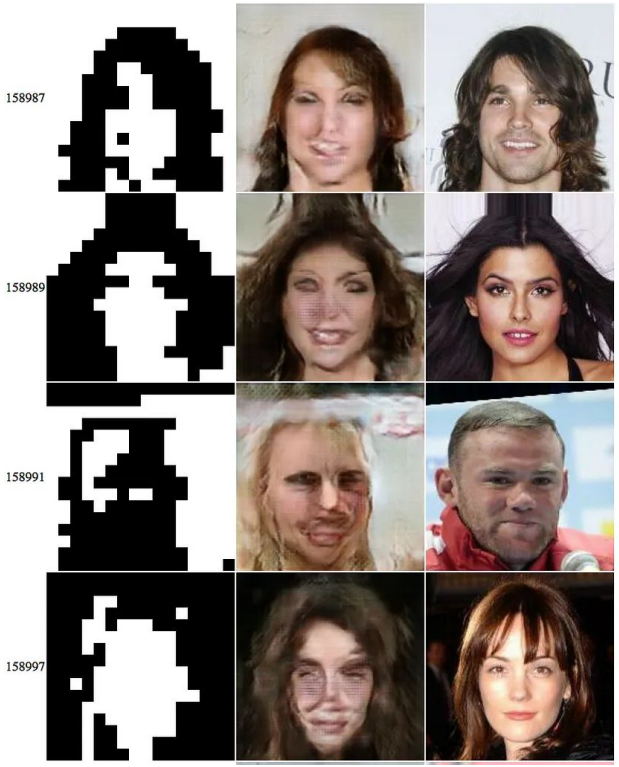
The black-box attack is general and applicable to other datasets, image distributions, and hash functions.

The image hash will be used as the input image to the Pix2Pix model.



Reference: https://affinelayer.com/pix2pix/

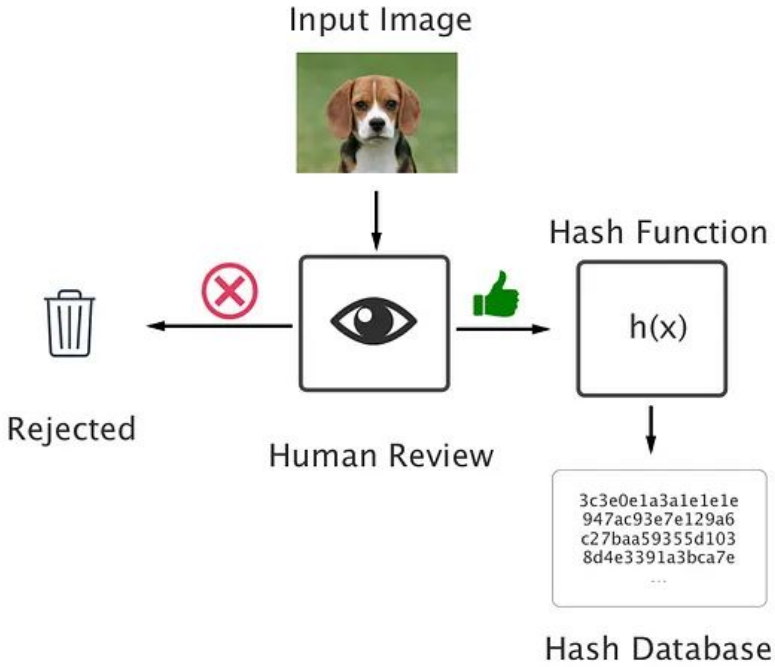
Black-box Attack: Hash Reversal



- In many instances, the attack can create a face that closely resembles the original.
- Even when it fails, it frequently still captures the correct gender, hairstyle, and race of the original image.

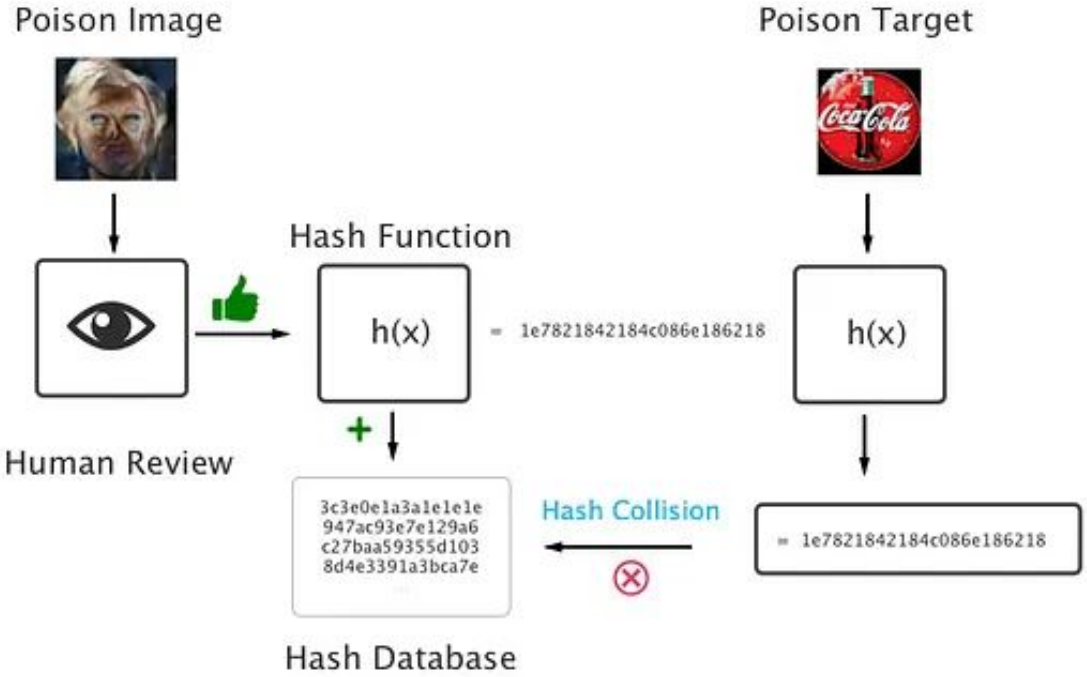
Reference: https://towardsdatascience.com/black-box-attacks-on-perceptual-image-hashes-with-gans-cc1be1

Black-box Attack: Hash Poisoning



Reference: https://towardsdatascience.com/black-box-attacks-on-perceptual-image-hashes-with-gans-cc1be11f277

Black-box Attack: Hash Poisoning



Reference: https://towardsdatascience.com/black-box-attacks-on-perceptual-image-hashes-with-gans-cc1be11f277

Black-box Attack: Hash Poisoning

What's the aim of the attacker?

- For the human reviewer to approve an image that has a hash collision with the Coca-Cola logo.

How to achieve that?

- Deceive a human reviewer to classify the Coca-Cola logo as a banned image → "poisoning" the database with hashes of images that should actually be allowed

This task of misleading the reviewer can be achieved using trained generative model.

Probabilistic Challenges due to Generative AI

Typical probabilistic strategies:

- Methods used to estimate the likelihood of different cybersecurity risks.
- E.g., determine the chance of a cyber attack happening vs the chance that an email is legitimate.
- Depend on the quality and amount of data available.
- Quality ↑, Prediction accuracy ↑

Does not include unpredictable human behaviors, like mistakes or being tricked (social engineering) → Generative AI makes it even harder to be detected!

Probabilistic Challenges due to Generative AI

Common Probabilistic Approaches:

- Phishable Authentication, i.e., passwords or one-time passwords (OTPs) → can be easily stolen or guessed
- Knowledge-Based Verification
- E.g., What was your first pet's name → These can sometimes be guessed or found out through social engineering
- Human-Verified Documentation:
- E.g., Using documents like passports for verification → can be forged with generative AI models.

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AI Defense Strategies

AI Attack

Common Tools for Detecting Deepfake

[Sentinel]
[Intel's FakeCatcher]
[WeVerify]
[Microsoft's Video Authenticator Tool]



Defense Strategies

1. Adversarial Training
- Inject adversarial examples into the training data
- Learn to recognize and handle these examples

2. Robust Architectures
- Develop neural network architectures that are inherently resistant to adversarial attacks.
- Incorporate additional layers and non-linear functions
- increase the difficulty for attackers in finding effective adversarial examples.

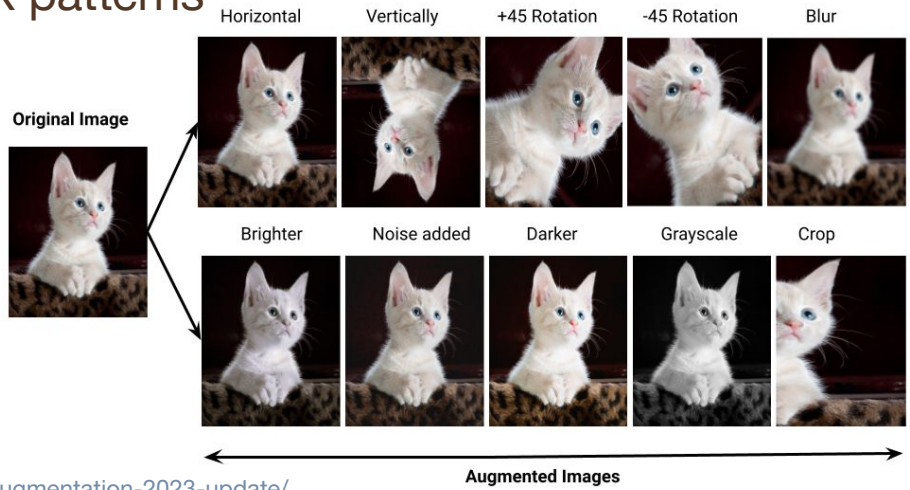
Defense Strategies

3. Ensemble Methods
   - Combine multiple models to enhance robustness.
   - Even if one model is vulnerable to adversarial attacks, the other models in the ensemble can contribute to more reliable decision-making.

4. Input Preprocessing
   - Transforming input data can enhance its resilience against adversarial perturbations.
   - E.g., adding noise or reducing dimensionality.

Defense Strategies

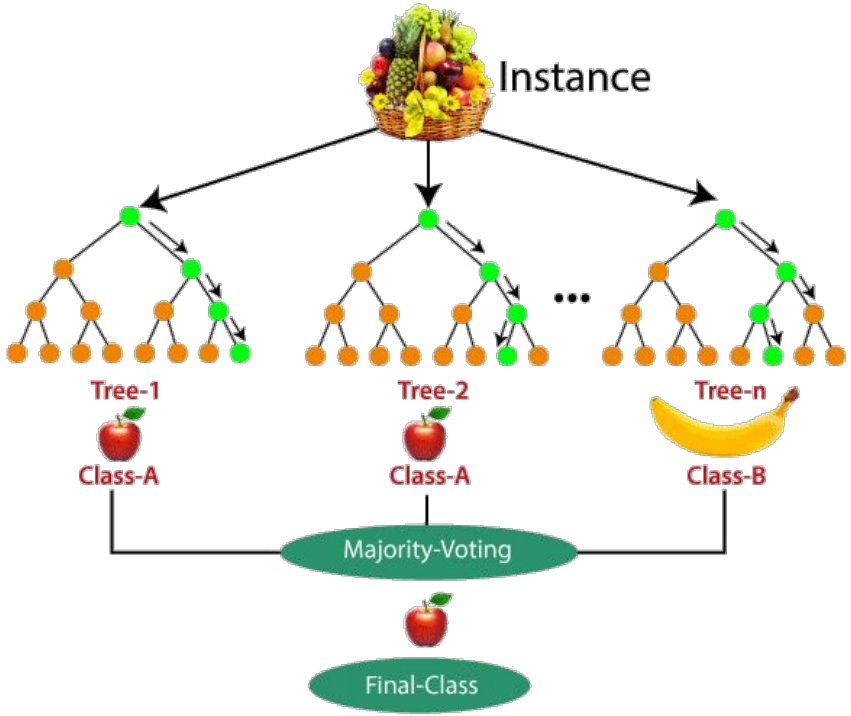
5. Data augmentation
   - Diversify your training data with random transformations and variations to improve model generalization
   - Reduce susceptibility to specific attack patterns
   - Prevent models from overfitting



Reference: https://ubiai.tools/what-are-the-advantages-anddisadvantages-of-data-augmentation-2023-update/

Ensemble Methods: Random Forest

- Random forest is used for both classification and regression tasks.
- Combines the outputs of multiple decision trees to improve the overall accuracy and reduce overfitting.

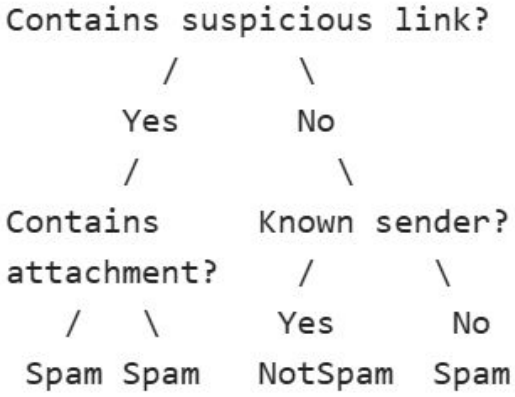


Reference: https://www.javatpoint.com/machine-learning-random-forest-algorithm

Ensemble Methods: Random Forest

Imagine you want to classify an email as Spam or Not Spam.

A decision tree asks a series of questions:



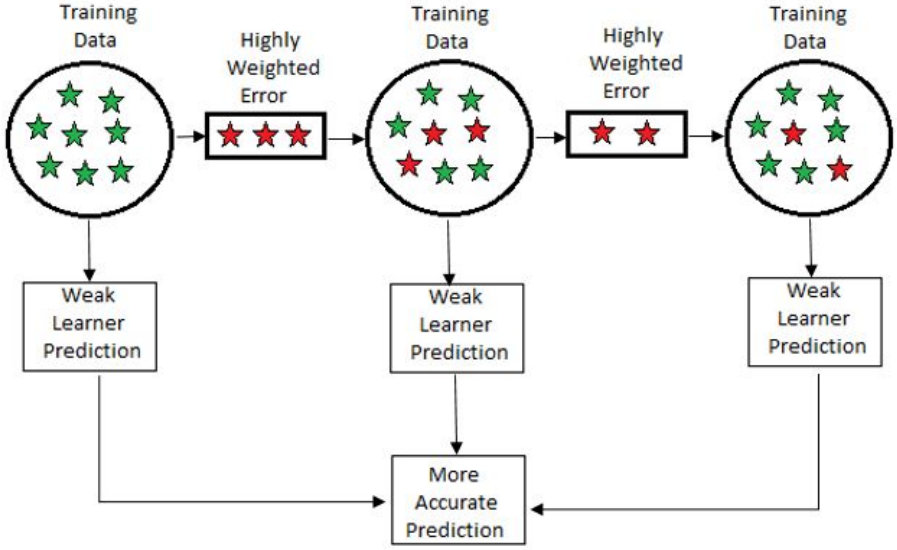
One tree makes one decision.

A single tree can be wrong.

Different tree make different mistakes, voting becomes more reliable.

Ensemble Methods: Gradient Boosting

- Gradient boosting builds a model sequentially where each new model corrects the errors of the previous ones.
- Improves predictions by combining the insights of numerous weak predictors to form a powerful and more accurate predictor
- E.g., XGBoost, LightGBM, and CatBoost



Reference: https://pub.towardsai.net/fully-explained-gradient-boosting-technique-in-supervised-learning-d3e293ca70e1

Ensemble Methods: Gradient Boosting

One student works on the problem:

```
Student 1
```

Gets some parts wrong.

A second student only looks at the mistakes:

```
Student 2 → fixes errors
```

A third student fixes the remaining errors:

```
Student 3 → fixes more errors
```

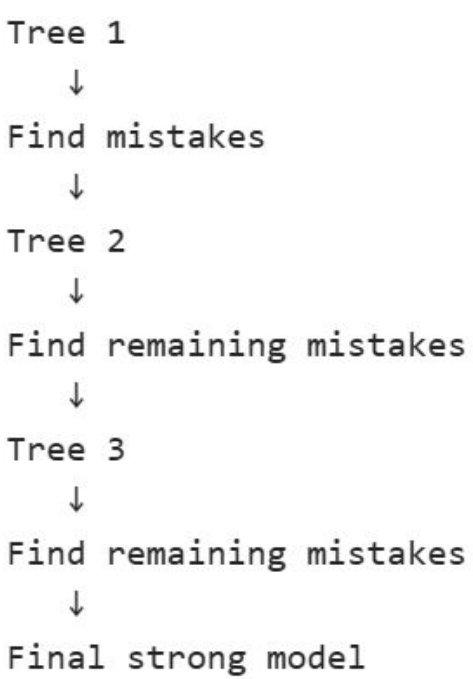
And so on.
This is why it is called **boosting**: each new model boosts the performance of the previous one.

Ensemble Methods: Gradient Boosting

Combine the predictions of the weak learner with the current model's predictions.

- Add a fraction (learning rate or shrinkage factor) of the weak learner's predictions to the current model's predictions.
- Helps the model move closer to the actual target values.

New model predictions = Current model predictions + (Learning rate * Weak learner predictions)



Random Forest vs Gradient Boosting

Random Forest	Gradient Boosting
Trees built independently	Trees built sequentially
Voting	Error correction
Less prone to overfitting	More prone to overfitting
Easier to train	More complex
Generally robust	Generally more accurate

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Differential Privacy Defense Strategy

Differential privacy

What is differential privacy?

- An algorithm is considered differentially private if the outcome statistics do not reveal any information about whether a specific individual was part of the dataset.
- Outcome statistics hardly changes when a single individual joins or leaves the dataset

How can that be done?

- Adding random noises to the original data
- The slightly-biased noise make individual users unidentifiable.
- The noises will average out over a large numbers of data points so valuable insights still can be obtained.

Differential privacy: RAPPOR

RAPPOR: Randomized Aggregatable Privacy-Preserving Ordinal Response

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ABSTRACT

Randomized Aggregatable Privacy-Preserving Ordinal Response, or RAPPOR, is a technology for crowdsourcing statistics from end-user client software, anonymously, with strong privacy guarantees. In short, RAPPORs allow the forest of client data to be studied, without permitting the possibility of looking at individual trees. By applying randomized response in a novel manner, RAPPOR provides the mecha-

asked to flip a fair coin, in secret, and answer "Yes" if it comes up heads, but tell the truth otherwise (if the coin comes up tails). Using this procedure, each respondent retains very strong deniability for any "Yes" answers, since such answers are most likely attributable to the coin coming up heads; as a refinement, respondents can also choose the untruthful answer by flipping another coin in secret, and get strong deniability for both "Yes" and "No" answers.

Surveys relying on randomized response enable easy com-

Differential privacy: RAPPOR

People tend to give false answers if being asked sensitive questions.

- Q: How to assure people to give reliable answers while giving them the chance to lie?
- Use Randomized Response technique → proposed by S. L. Warner in 1965

In such a process, chances decide whether the question is to be answered truthfully, or “yes”, regardless of the truth.

- This chance is controlled by the respondent
- Unknown to the interviewer

Differential privacy: RAPPOR

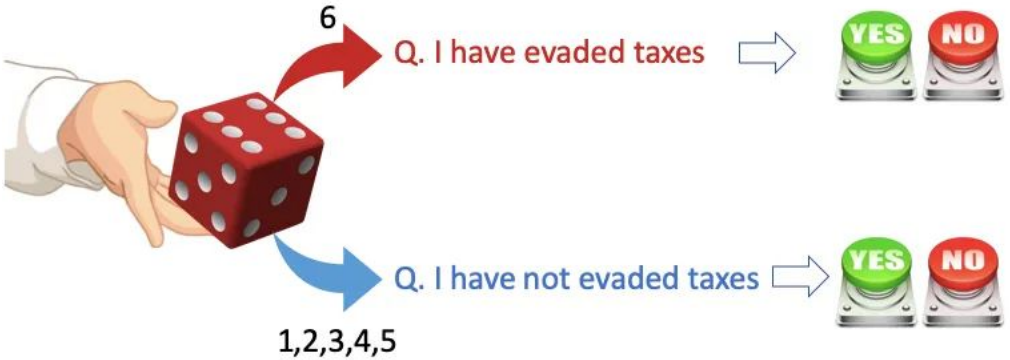
Example of Randomized Response Technique:

- Suppose a social scientist needs to know the percentage of people who have ever evaded taxes. To get the outcome statistics, the social scientist can survey 1,000 people randomly and count how many answer “Yes”.
- But many respondents will not give their true answers. They may fear this sensitive question will get them into trouble.
- Each respondent rolls the dice without letting the interviewer know the outcome.

Differential privacy: RAPPOR

If it is a 6, answer “Yes” or “No” to the first question (Q1), otherwise answer the opposite question (Q2):

- Q1. I have evaded taxes
- Q2. I have never evaded taxes.



Reference: https://medium.com/analytics-vidhya/what-is-differential-privacy-553f41a757fd

Differential privacy: RAPPOR

To get the true answer, let:

- T = the truth percentage that the researcher wants to know
- p = the probability → 1/6
- S = the surveyed percentage known to the researcher

The surveyed percentage S can be expressed as:

$$S = p * T (1 - p) * (1 - T)$$

After arranging the terms, the true percentage T can be computed as:

$$T = (S + p - 1) / (2p - 1)$$

Differential privacy: RAPPOR

There is an issue with the typical Randomized Response Technique:

- If the one-time randomized response is done on the same person repeatedly, the sequence of data of an individual still can be used to reveal his identity → **longitudinal** privacy

Permanent Randomized Response

- Replaces the one-time randomized response with a randomized noisy value permanently
- All future data will use the same derived randomized value.
  - E.g., The derived noise can be ‘1’, ‘0’, or the original one-time randomized response value with certainly probabilities respectively (summing up to 1).
  - Ensure privacy because the attackers can not differentiate between true and “noisy” data in the longitudinal data.

Differential privacy: RAPPOR

Example of Permanent Randomized Response:

- Suppose each individual have to report their age (in days) everyday. After a period of time, some individual's data will stop being collected, or mixed with an increased noises permanently.

This makes the longitudinal data of the individual unidentifiable by attackers.

Differential Privacy for Synthetic Data Generation

Generative Large Language Models (LLMs) can create synthetic text by sampling their outputs.

- Can be fine-tuned using relevant data, i.e., scientific papers and produce authentic scientific writing.

What's the catch?

- Generating synthetic text from private documents, carries privacy risks → LLMs can potentially memorize and reproduce training data.

Differential Privacy could be the solution.

Adversarial AI Simulation

Attack Scenario

Choose ONE scenario below:

1. AI Phishing Email: Generate a hyper-personalised phishing message using AI.
2. Vishing with Deepfake Voice: Design a scenario where a cloned voice is used in fraud.
3. Black-box Hash Attack: Craft a poisoning/reversal attack using perceptual hashes.
4. LLM Prompt Injection: Design a malicious prompt that forces a model to reveal hidden data.

Share your answer here: https://shorturl.at/geiwv

Adversarial AI Simulation

- Craft adversarial strategies.
- Propose countermeasures.

In your slide, include:

- What method you used, how it works, and its potential impact.
- How you would mitigate it, including advantages and limitations.

Share your answer here: https://shorturl.at/geiwv

Scenario

Scenario:
In January 2024, thousands of AI-generated robocalls impersonating President Joe Biden went out to Democratic voters discouraging them from voting in the New Hampshire primary.

Q1: Describe how does the deepfake use impact the integrity of elections?

Q2: How can we combat this?

Share your answer here: https://shorturl.at/geiwv